

AGHA ABRAR KHAN

Full Stack AI Engineer | Python & TypeScript | Generative AI & RAG

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PROFESSIONAL SUMMARY

Full Stack ML Engineer with 4+ years of freelance and professional software delivery experience, combining full-stack development with hands-on work in machine learning, generative AI, and intelligent automation. Skilled in building responsive, API-driven web applications using Python, FastAPI, JavaScript, TypeScript, React, Next.js, Node.js, and SQL/NoSQL databases. Experienced in integrating AI models into applications, developing backend APIs, and connecting data workflows with interactive user interfaces. Built RAG systems, LLM-powered assistants, and computer-vision applications using OpenAI APIs, Hugging Face, LangChain, embeddings, and vector search. Strong problem solver with experience delivering projects from requirements and architecture through integration, testing, deployment, and documentation.

TECHNICAL SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Tailwind CSS, Responsive UI, Component Architecture, Dashboard Development, API-Driven Interfaces, State Management

Backend & APIs: Node.js, Express.js, Python, FastAPI, Flask, REST APIs, Webhooks, Microservices, API Design, API Integration, Server-Sent Events, WebSockets

Databases & ORMs: MySQL, Prisma, PostgreSQL, Supabase, MongoDB, SQL, Relational Schema Design, Data Modeling, Query Optimization Basics

Modern TypeScript Stack: TypeScript, Tailwind CSS, Prisma, MySQL, tRPC Familiarity, Type-Safe API Patterns, Full-Stack Product Architecture, Reusable Code Practices

Security & Fintech Basics: Secure API Practices, Authentication Concepts, Authorization/RBAC, Input Validation, Data Integrity, Sensitive Data Handling, Audit Logging Basics

AI & Automation: OpenAI API, Hugging Face, Ollama, LangChain, LangGraph, RAG, AI Chatbots, AI Copilots, Prompt Engineering, Custom Backend Automation

Cloud, DevOps & Tools: Docker, Git, GitHub, Vercel, Supabase, AWS/S3 Basics, CI/CD Familiarity, Environment Configuration, Deployment Troubleshooting

Collaboration: Requirements Understanding, Solution Architecture, Product Collaboration, Documentation, Agile Delivery, Code Reviews, Stakeholder Communication

EDUCATION

Sukkur IBA University

BS Computer Science (BSCS)

Sukkur, Pakistan

PROFESSIONAL EXPERIENCE

Sukkur IBA University

Instructor / Visiting Faculty

September 2025 – May 2026

Sukkur, Pakistan

- Taught Web Development and Java Development courses under the Peoples Information Technology Program (PITP), Phase-II, at Sukkur IBA University.

Prime Institute of Science & Technology

Instructor

2020 – 2022

Sukkur, Pakistan

- Taught various courses during two years as an instructor.

Freelancing

Freelance Full Stack & AI Developer

2023 – 2026

Remote

- Developed full-stack web applications, dashboards, and AI-enabled solutions using React, Next.js, Python, Flask, FastAPI, and REST APIs.
- Built backend services, integrated databases and third-party APIs, and implemented automation workflows based on client requirements.
- Managed projects from requirements gathering and development through testing, deployment, documentation, and client handover.

KEY PROJECTS

- MCP-Powered GitHub Repository Assistant:** Built a Next.js and TypeScript application integrating OpenAI and GitHub REST APIs through an MCP-style tool layer to analyze repository activity, prioritize issues, summarize pull requests, identify documentation gaps, and generate sprint plans.
- AI-Powered Clinical Decision Support System:** Developed an educational prototype using React, TypeScript, NestJS, FastAPI, Prisma, and PostgreSQL, with code for patient management, role-based access, audit logging, deterministic medication-safety checks, and an LLM gateway with de-identification and structured response validation.
- Resume Ranking System:** Built a Python and Flask application that extracts text from uploaded PDF resumes, preprocesses content, and ranks resumes against job descriptions using TF-IDF vectorization and cosine similarity, presenting results through a web dashboard.
- Transparent Auction House – Final Year Project:** Developed a React and Flask auction application with JWT-protected bidding APIs and bid-history tracking. Implemented bidding-behavior feature extraction and a model-inference module for detecting suspected shill bidding.
- Agentic AI RAG System:** Built a multi-agent Retrieval-Augmented Generation pipeline using LangChain and vector databases, incorporating retrieval reasoning, self-correction, and grounded response generation from private documents.

CERTIFICATIONS

- **Claude with Amazon Bedrock** – Anthropic Education
- **MCP: Advanced Topics** – Anthropic Academy
- **Python and Flask Framework Complete Course** – Udemy
- **Generative AI and Artificial Intelligence (AI) for Leaders** – Udemy
- **Complete JavaScript with HTML5, CSS3 from Zero to Expert – 2026** – Udemy
- **NodeJS Masterclass (Express, MongoDB, OpenAI)** – Udemy
- **Flutter Masterclass (Dart, API & More)** – Udemy
- **Diploma in Information Technology (DIT)** – Trade Testing Board Sindh
- **Certificate in Information Technology (CIT)** – Prime Institute of Science & Technology, Sukkur
- **English Language Diploma** – Prime Institute of Science & Technology and English Language, Sukkur

LANGUAGES

English, Urdu, Sindhi